TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Jack Bloem	Email	jack.bloem@tno.nl
Institute Name	Netherlands Organisation for applied scientific research, TNO			
Address	Anna van Buerenplein 1, The Hague, The Netherlands			

2. Technology Overview

Title of Proposed Technology	An Engine Aftertreatment Technology that allows emission reduction for sustainable fuels in order to meet Stage V, Stage VI (future) and Euro 7			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical materials		Energy & Resources	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile / Rail Vehicles	Engines and Power Transmission Systems	New and Renewable Energy	Bio-fuels
Concept of Proposed Technology	We apply model based aftertreatment controls to optimized performance over the complete lifetime of an engine aftertreatment system (1),(2),(3),(4). The models are physics based. The models are ECU embeddable, hence they are applied in real time. The concept consists of one model for both the development of the system as well as the application in real time controls. This technology is an enabler for engines that run on sustainable fuels, to make them compliant with Stage V, VI and Euro 6/7 regulation. *Describe the core concept, main functions, and potential uses of the proposed technology*			
Rationale for Joint R&D Project	Future engine systems are becoming more and more complex. This technology reduces development times, hence cost. Models can for example be used for off-line calibration (5), reducing engine test cell time and costs. And because the concept is model based, the system will be more robust to real world disturbances. *Describe the need for this joint R&D project and why collaboration is required*			

Key Words	1. model based aftertreatment controls 2. ECU embeddable 3. Stage V, Stage VI and Euro 7 4. Automotive, Non-road mobile machinery and Maritime 5. *Provide key terms that describe the proposed technology and its main features (min 3, max 5)*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	R&D objective is twofold 1. Demonstrate that this technology concept works on the hardware for our Korean partner 2. Demonstrate that this technology concept works for sustainable fuels (hydrogen, methanol, bio fuels) *Describe the specific technologies you aim to develop through this R&D project*			
R&D Target Specifications	Achieve Stage V, VI (future) or Euro 7 legislative requirements. *Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters*			
R&D Plan	Total project duration 18 months. Project consists of following work packages: WP1 Virtual Exhaust Line calibration and validation (months 1 to 8) Specification of catalyst components required for model calibration. Korean partner provides aftertreatment component hardware to TNO. TNO will built-up aftertreatment model calibration setup, execute the measurements and calibrates and validates the aftertreatment models. WP2 Model-based controls: calibration and validation (months 9 – 15) Models will be used to off-line calibrate the dosing controls for engine/aftetreatment application of Korean partner. TNO will embed the calibrated models on rapid control prototyping system and demonstrates the performance in TNO test cell. Korean partner invited on-site. WP3 Knowledge transfer workshops (months 15 – 18) TNO will provide software under license. Hands-on workshop on models, model calibration and aftertreatment controls. Workshop material will be shared. *Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives*			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Model & control development • Demonstration of the technology concept • Training (to teach the complete methodology and to teach the Korean partner to apply the technology themselves) *Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.*
Expected R&R of Korean SME Partner	• Provides the engine and aftertreatment hardware • TNO ships the demonstrator to Korea and Korean partner verifies the performance in their own facility • Transition of the TNO demo to a target control platform, fit for production *Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The results can be applied in automotive, Non-Road Mobile Machinery, and in Maritime sectors. The outcome is a demonstrator that can be further engineered for TRL levels 6 and higher. Also an outcome is that our Korean partner is able to apply this technology themselves for future versions of new hardware. The impact is that the partner is qualified to sell Stage V, VI or Euro 7 compliant after treatment products, for sustainable fuels. They will contribute in this way to a cleaner environment, with less emission of CO2 and other harmful emissions. *Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.*

6. References

(1) Cloudt, R., Saenen, J., van den Eijnden E., Rojer, C., Virtual Exhaust Line for Model-based Diesel Aftertreatment Development, SAE Technical paper, 2020-01-0888, 2010

(2) Ramachandran, S., Hommen, G., Mentink P., Seykens X., et al. Robust Cost-Optimal and Compliant Engine and Aftertreatment Operation using Air-path Control and Tailpipe Emission Feedback, SAE Int. J. Engines 9(3):2016, doi:10.4271/2016-01-0961

(3) Hommen, G., Kupper, F., and Seykens, X., "Robust, Model-Based Urea Dosing Control for SCR Aftertreatment Systems using a Cross-Sensitive Tailpipe NOx Sensor," SAE Technical Paper 2017-01-0938, 2017, doi:10.4271/2017-01-0938.

(4) Seykens, X., Kupper, F., Mentink, P., and Ramesh, S., "Towards Ultra-Low NOx Emissions within GHG Phase 2 Constraints: Main Challenges and Technology Directions," SAE Technical Paper 2018-01-0331, 2018, https://doi.org/10.4271/2018-01-0331.

(5) Straten, T., and van den Berk, J., "Model-Based Approach for Calibration and Validation by Simulation of Emission Control Solutions for Next Generation Off-Road Vehicles," SAE Technical Paper 2011-01-0309, 2011, https://doi.org/10.4271/2011-01-0309.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization